Helix -- Executive summary

Helix is an Intelligent SDLC Copilot that converts unstructured requirements into traceable stories, tasks, acceptance criteria, tests, ambiguity findings, risks, and effort estimates. Teams review and refine work in a React workspace, query a copilot over project context, and export to CSV/Markdown and optionally Jira or GitHub.

What was built (hackathon scope)

Requirement ingestion (text, file, URL, optional browser voice-to-text), preprocessing and clause-level traceability, multi-agent AI generation with live SSE progress, interactive dashboard (Kanban, summary, readiness, tests, ambiguity, export), test-case generation, ambiguity detection, conversational assistant, effort metrics, and governed export with human approval flags.

Technology

Frontend: React + Vite (helix-frontend). Backend: Python 3.11 + FastAPI + SQLAlchemy; PostgreSQL in Docker Compose; optional Redis, MongoDB, FAISS RAG; Azure OpenAI and/or Anthropic when keys are configured, with a mock path for offline demos.

How to verify quickly

Sign in with the seeded demo account (see README / docs/RUNBOOK.md). New Project: Load sample requirement, Ingest, then Generate artifacts -- no microphone required. Full stack: docker compose up --build from repo root, or deploy single-container demo per docs/DEMO_HOSTING.md.

Repository

https://github.com/siddham04/AIThon -- see docs/IMPLEMENTATION_REPORT.md for the full written report and docs/RUNBOOK.md for reviewer steps.

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Logical flow (high level)
  Browser (React) --> FastAPI (/api/*)
       |-> PostgreSQL / SQLite
       |-> optional Redis, MongoDB
       |-> RAG (FAISS) + Azure / Anthropic / mock agents
```